Practical 10

Aim:-   Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

Code:-

```
  GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>

int main() {
    int req[] = {98,183,41,122,14,124,65,67};
    int n = 8;
    int head = 53;
    int visited[8] = {0};
    int total_movement = 0;

    int i, j, min_dist, index;

    for(i=0;i<n;i++) {
        min_dist = 1000;

        for(j=0;j<n;j++) {
            if(!visited[j]) {
                int dist = abs(head - req[j]);

                if(dist < min_dist) {
                    min_dist = dist;
                    index = j;
                }
            }
        }

        total_movement += min_dist;
        head = req[index];
        visited[index] = 1;
    }

    printf("\nTotal Head Movement = %d\n", total_movement);

    return 0;
}
```

Output:-

```
Total Head Movement = 323
```